

Product Specification: "Executive Summary" Canvas Tool

1. Executive Overview

1.1 Purpose & Value Proposition

Enterprise Architects (EAs) often struggle to translate complex ArchiMate models and lengthy technical architecture documents into concise, high-impact presentation decks suitable for C-Level executives (CIO, CTO, CEO, Board Members).

The **"Executive Summary" Canvas Tool** automates and streamlines this translation process. By analyzing technical models and documentation through an intelligent multi-agent pipeline, the tool extracts core strategic concepts, synthesizes C-Level key messages, establishes a coherent narrative thread ("storyboard"), and builds a polished presentation deck. Crucially, it combines automated AI synthesis with human-in-the-loop control, allowing the architect to refine the narrative before final slide rendering.

1.2 Target Audience

- **Primary:** Enterprise Architects, Solution Architects, Domain Architects.
- **End Audience (Consumers):** C-Level Executives, Strategic Steering Committees, IT Directors, Board Members.

2. System Architecture & High-Level Workflow

The tool operates in four discrete phases:

```
[Phase 1: Ingestion & Extraction]
  |   • Parse ArchiMate Models (XML/JSON) & Architecture Docs
  |   (PDF/DOCX/MD)
  ▼
[Phase 2: LangGraph Narrative Synthesis]
  |   • Multi-agent analysis (Concepts -> C-Level Insights ->
  |   Narrative Thread)
  ▼
[Phase 3: Human-in-the-Loop Canvas Refinement]
  |   • EA adjusts thread, adds/removes concepts, reorganizes
  |   slide sequence
  ▼
[Phase 4: Visual Generation & Collabora Integration]
  |   • Image extraction from source docs
  |   • Dynamic diagram synthesis (ArchiMate/UML viewports)
```

- Collabora presentation deck generation (.pptx/.odp)

3. Functional Specifications

FS-1: Source Linking & Document Ingestion

- **FS-1.1 Input Formats:**
 - **Architecture Models:** ArchiMate Open Exchange Format (.xml), Archi models, JSON exports.
 - **Documents:** PDF, Microsoft Word (.docx), Markdown (.md), HTML, Powerpoint, etc..
- **FS-1.2 Ingestion & Indexing:**
 - The ingestion is already done by the Semantic Canvas.
 - Linked documents are already vectorized.

FS-2: LangGraph Analysis & Concept Extraction

- **FS-2.1 Concept Mining:**
 - Identify high-level architectural drivers, constraints, capabilities, and target state transitions.
 - Filter out low-level implementation details not relevant to executive stakeholders.
- **FS-2.2 Executive Messaging:**
 - Synthesize technical concepts into strategic impact statements (e.g., Cost, Risk, Time-to-Market, Compliance, Business Capability Enablement).
- **FS-2.3 "Common Thread" (Storyboard) Generation:**
 - Generate a structured, slide-by-slide outline representing a logical narrative flow (e.g., *Current State Pain Points* → *Strategic Vision* → *Target Architecture* → *Business Value* → *Roadmap & Next Steps*).
 - For each slide, define:
 - Slide Title & Executive Takeaway (1 line).
 - Core Concept bullet points.
 - Visual Asset Strategy (Extract existing image vs. Generate new architecture/UML diagram vs. Text layout).

FS-3: Interactive Canvas & Narrative Refinement (Human-in-the-Loop)

- **FS-3.1 Narrative Storyboard View:**
 - A visual node/card layout representing each slide in the story thread.
- **FS-3.2 Editing Capabilities:**
 - **Reorder/Delete/Add Slides:** Move slide cards around or insert custom slides.
 - **Concept Management:** Drag-and-drop extracted concepts between slides or dismiss irrelevant concepts.
 - **Prompt Tuning:** Allow the user to provide micro-prompts (e.g., "Make Slide 3 focus more on cybersecurity compliance").
- **FS-3.3 Live Concept Map:**

- An interactive graph displaying extracted ArchiMate elements and how they map to individual slides.

FS-4: Visual Asset Mining & Dynamic Diagram Synthesis

- **FS-4.1 Image Extraction:**
 - Locate and extract embedded visual diagrams/images from source documents.
 - Perform OCR/Caption analysis to match images to the relevant slide context.
- **FS-4.2 Automated Diagram Generation:**
 - When an abstract concept or architecture relationship lacks an existing figure, the tool generates clean visual diagrams.
 - **Supported Diagram Types:** High-level ArchiMate Viewpoints (Layered, Application Cooperation, Strategic Capability Map), Sequence Diagrams, UML Class/Component, C4 Models.
 - **Rendering Engines:** PlantUML, Mermaid.js, or Graphviz converted into clean vector SVG/PNG graphics styled with enterprise theme palettes.

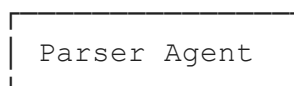
FS-5: Presentation Generation via Collabora (or alternative - To Be Defined)

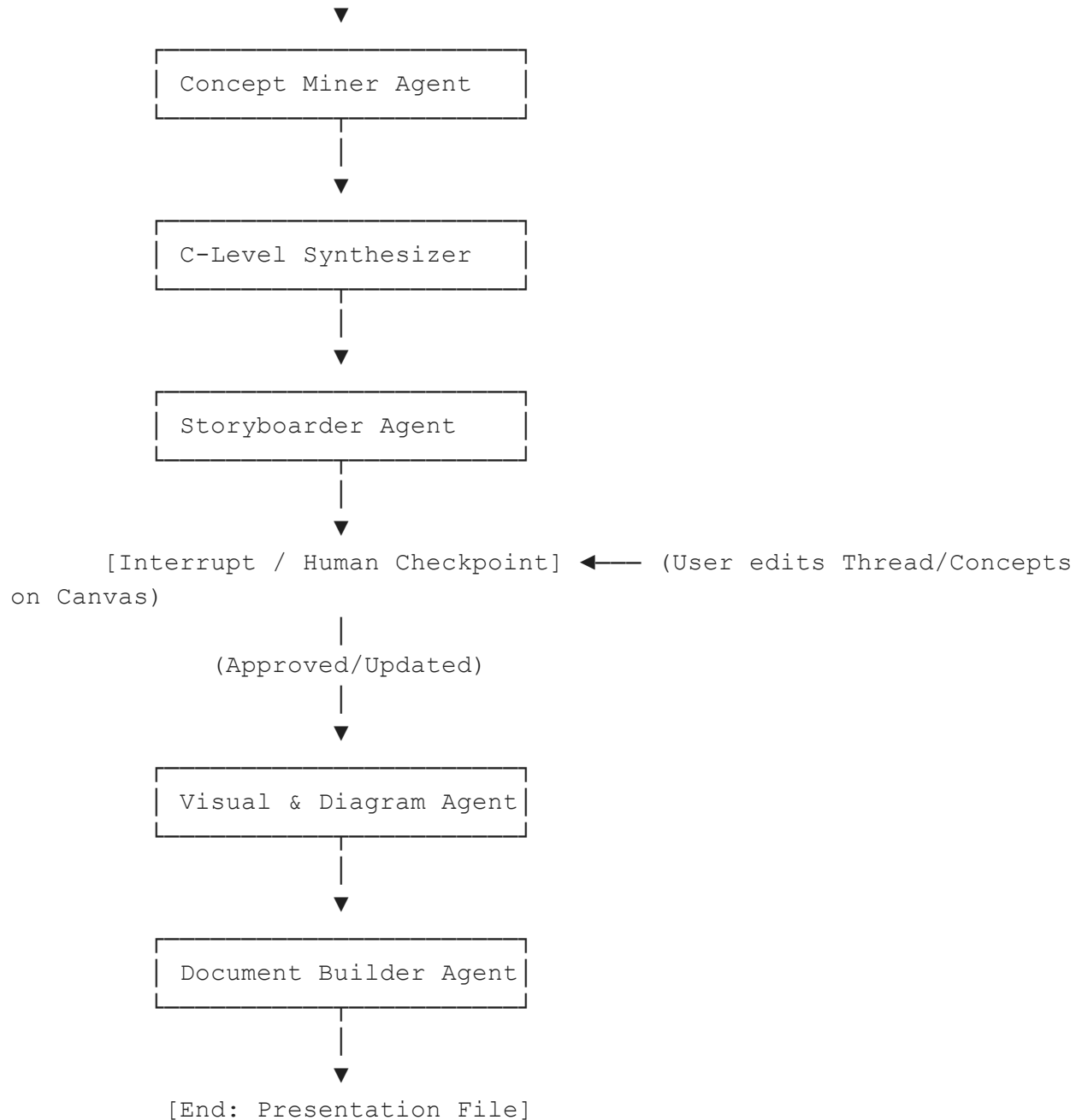
- **FS-5.1 Collabora Online Integration:**
 - Utilize Collabora Online API / LibreOffice technology engine to assemble presentation files (.pptx / .odp).
- **FS-5.2 Branding & Template Application:**
 - Apply standard enterprise PowerPoint/Collabora templates (Master Slides, standard layouts, color palettes, typography).
- **FS-5.3 Asset Injection:**
 - Embed generated text, bullet points, extracted source images, and newly synthesized diagrams directly into template placeholder frames.
- **FS-5.4 Interactive Editing in Collabora:**
 - Upon generation, open the resulting presentation directly within an embedded Collabora Online frame inside the application for final manual polish.

4. LangGraph Agentic Architecture

The AI engine uses a stateful, multi-agent workflow powered by **LangGraph** to ensure deterministic, traceable, and inspectable synthesis.

[Start: Linked Docs & Models]





4.1 Agent Roles & Responsibilities

1. Parser Agent:

- Extracts ArchiMate XML structure (elements, relations, views) and text/media chunks from linked documents.

2. Concept Miner Agent:

- Group elements into logical themes (e.g., Cloud Migration, Legacy Systems, Capability Gaps).

- 3. **C-Level Synthesizer Agent:**
 - Translates technical clusters into executive outcomes (ROI, Risk, Speed, Compliance).
- 4. **Storyboarder Agent:**
 - Organizes synthesized takeaways into a coherent presentation structure ("Common Thread").
- 5. **Visual & Diagram Agent:**
 - Determines visual needs for each slide. Generates PlantUML/Mermaid markup for missing diagrams and crops/formats extracted pictures.
- 6. **Document Builder Agent:**
 - Prepares payload for Collabora API, populating layout templates with formatted text and media.

5. User Interface (Canvas Tool) Layout

The Canvas interface is divided into three primary regions:

Header: Project Name Source Documents (3) Model Loaded [Generate Slide Deck]	
Left Panel: Concept Graph • ArchiMate Capabilities • Business Drivers • Target Architecture Nodes • Extracted Figures (Gallery)	Main Canvas: Narrative Thread Storyboard Slide 1: Executive Overview [Takeaway] Accelerating Cloud Transition [Concepts] #Migration #CostOptimization ↓ Slide 2: Target Capability Architecture [Takeaway] API-First Platform Integration [Diagram Preview] (ArchiMate View)

- 1. **Left Panel (Asset & Concept Explorer):** Shows all auto-extracted concepts, ArchiMate nodes, and doc images. Allows manual drag-and-drop onto specific slides.
- 2. **Central Canvas (Common Thread Storyboard):** Interactive cards representing the slides in sequence. Users edit text, change slide order, adjust visual strategy, or re-run AI generation on specific slides.
- 3. **Collabora Embedded Viewer (Post-Generation):** Once the user clicks "Generate Slides", the UI seamlessly embeds Collabora Online to edit and fine-tune the final deck.

7. Non-Functional Requirements & Governance

- **Enterprise Security & Privacy:** Architectural models contain sensitive strategy details. All processing must support on-premise or private cloud deployment (local LLM/LangGraph execution).
- **Traceability:** Every generated slide takeaway must maintain a traceable link back to the source document paragraph or ArchiMate model ID.
- **Performance:**
 - Initial ingestion & narrative generation: < 45 seconds for standard models/docs.
 - Slide deck rendering: < 15 seconds.